

Lecture 3: Torsion and Frenet curves

September 10

Let $c = c(s)$ be a unit-speed curve in $\mathbb{R}^3$ with

$$\kappa(s) > 0.$$

From Lecture 2,

$$T = c', \quad T' = \kappa N.$$

Definition. The *binormal vector* is

$$B = T \times N.$$

The ordered triple (T, N, B) is a positively oriented orthonormal basis at every point of the curve. It is called the *Frenet frame*.

The plane spanned by T and N is the osculating plane. The vector B is perpendicular to this plane.

Since B is a unit vector,

$$B' \perp B.$$

Since $B \perp T$, differentiation gives

$$\langle B', T \rangle = -\langle B, T' \rangle = -\kappa \langle B, N \rangle = 0.$$

Therefore B' is parallel to N .

Definition. The *torsion* of the curve is the function τ defined by

$$B' = -\tau N.$$

Torsion measures the rate at which the osculating plane rotates. It measures the failure of the curve to remain in a plane.

The derivatives of the Frenet frame are determined by κ and τ .

We already know

$$T' = \kappa N, \quad B' = -\tau N.$$

To find N' , use $N = B \times T$:

$$N' = B' \times T + B \times T' = (-\tau N) \times T + B \times (\kappa N).$$

Hence

$$N' = -\kappa T + \tau B.$$

Thus the *Frenet equations* in $\mathbb{R}^3$ are

$\begin{aligned} T' &= \kappa N, \\ N' &= -\kappa T + \tau B, \\ B' &= -\tau N. \end{aligned}$
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For an arbitrary regular parameter t , torsion can be computed without first passing to arclength:

$$\tau(t) = \frac{\det(c'(t), c''(t), c'''(t))}{\|c'(t) \times c''(t)\|^2}.$$

This formula is valid wherever

$$c'(t) \times c''(t) \neq 0,$$

or equivalently wherever $\kappa(t) > 0$.

The numerator is the scalar triple product

$$\langle c' \times c'', c''' \rangle.$$

It detects the component of c''' perpendicular to the osculating plane.

Curvature is already visible in the first two derivatives. Torsion first appears in the third derivative.

For a unit-speed curve,

$$c'' = \kappa N.$$

Differentiating and using the Frenet equations gives

$$c''' = -\kappa^2 T + \kappa' N + \kappa \tau B.$$

Therefore

$$\begin{aligned} c(s+h) &= c(s) + hT + \frac{h^2}{2}\kappa N \\ &\quad + \frac{h^3}{6}(-\kappa^2 T + \kappa' N + \kappa \tau B) + O(h^4). \end{aligned}$$

The curvature controls the second-order bending in the osculating plane. The torsion controls the first motion out of that plane, which occurs at third order.

Definition. A curve in $\mathbb{R}^3$ is ***planar*** if its image is contained in an affine plane.

Theorem. Let c be a unit-speed curve with $\kappa > 0$. Then c is planar if and only if

$$\tau = 0.$$

Sketch of proof. If the curve lies in a plane, its binormal is a constant unit normal to that plane. Thus $B' = 0$, so $\tau = 0$.

Conversely, if $\tau = 0$, then

$$B' = -\tau N = 0.$$

Thus B is constant. Since $\langle c', B \rangle = 0$, the function

$$\langle c(s), B \rangle$$

is constant. Hence the image of c lies in a plane perpendicular to B .

Definition. A curve $c : I \rightarrow \mathbb{R}^3$ is a **Frenet curve** if it is regular and its curvature never vanishes.

After an arclength reparametrization, the vectors

$$T, \quad N, \quad B$$

are then defined smoothly along the entire curve.

The hypothesis $\kappa > 0$ is important. At a point where $\kappa = 0$, the vector

$$N = \frac{T'}{\|T'\|}$$

is not defined, and consequently neither are B and τ .

A straight line is regular but is not a Frenet curve in this sense.

Definition. A Frenet curve $c = c(s)$ in $\mathbb{R}^3$ is a ***slope line*** if there is a fixed unit vector a such that

$$\langle T(s), a \rangle$$

is constant.

Equivalently, every tangent line makes the same angle with the fixed direction a . Slope lines are also called general helices.

Theorem of Lancret. A Frenet curve in $\mathbb{R}^3$ is a slope line if and only if

$$\frac{\tau}{\kappa}$$

is constant.

Thus neither constant curvature nor constant torsion is required. Only their ratio must remain constant.

Proof, one direction. Suppose $\langle T, a \rangle$ is constant. Differentiating gives

$$0 = \langle T', a \rangle = \kappa \langle N, a \rangle.$$

Since $\kappa > 0$,

$$\langle N, a \rangle = 0.$$

Hence a lies in the plane spanned by T and B :

$$a = uT + vB.$$

The quantity $u = \langle a, T \rangle$ is constant by assumption. Also,

$$v' = \langle a, B' \rangle = -\tau \langle a, N \rangle = 0,$$

so $v = \langle a, B \rangle$ is constant. Differentiating a gives

$$0 = a' = (u\kappa - v\tau)N.$$

Therefore

$$\frac{\tau}{\kappa} = \frac{u}{v},$$

with the case $v = 0$ excluded by $\kappa > 0$.

Proof, the converse. Suppose

$$\frac{\tau}{\kappa} = m$$

is constant. Consider the vector field

$$A = mT + B.$$

Using the Frenet equations,

$$A' = m\kappa N - \tau N = 0.$$

Thus A is a fixed vector. Moreover,

$$\langle T, A \rangle = m \quad \text{and} \quad \|A\| = \sqrt{1 + m^2}$$

are constant.

It follows that T makes a constant angle with the fixed unit vector

$$a = \frac{A}{\|A\|}.$$

Therefore c is a slope line.

Consider the circular helix

$$c(s) = \left(\frac{\kappa}{\kappa^2 + \tau^2} \cos(\sqrt{\kappa^2 + \tau^2} s), \frac{\kappa}{\kappa^2 + \tau^2} \sin(\sqrt{\kappa^2 + \tau^2} s), \frac{\tau}{\sqrt{\kappa^2 + \tau^2}} s \right),$$

where $\kappa > 0$ and τ are constants.

This curve has unit speed, curvature κ , and torsion τ . Its tangent makes a constant angle with the vertical axis because

$$\langle T, (0, 0, 1) \rangle = \frac{\tau}{\sqrt{\kappa^2 + \tau^2}}.$$

When $\tau = 0$, the helix becomes a circle of radius $1/\kappa$.

These examples suggest that curvature alone does not determine a space curve.

A circle and a circular helix can have the same constant curvature, but their torsions are different:

	κ	τ
circle	κ_0	0
helix	κ_0	τ_0

The curvature records bending inside the osculating plane. The torsion records how that plane twists through space.

Together, κ and τ should determine the shape of a Frenet curve in $\mathbb{R}^3$. Its position and orientation in space must still be specified separately.

Let

$$F(x) = Ax + b, \quad A \in SO(3),$$

be an orientation-preserving Euclidean motion. If $\widetilde{c} = F \circ c$, then

$$\widetilde{\kappa} = \kappa, \quad \widetilde{\tau} = \tau.$$

Translations and rotations alter the position of a curve, but not its curvature or torsion.

Consequently, no theorem can recover a unique curve from κ and τ alone. The best possible conclusion is uniqueness up to a Euclidean motion that preserves orientation.

Alternatively, uniqueness can be obtained by fixing one point of the curve and one positively oriented orthonormal frame at that point.

The same construction works in $\mathbb{R}^n$.

Definition. A unit-speed curve $c : I \rightarrow \mathbb{R}^n$ is a Frenet curve if

$$c'(s), c''(s), \dots, c^{(n-1)}(s)$$

are linearly independent for every $s \in I$.

Apply the Gram–Schmidt process to these vectors. This produces orthonormal vectors

$$e_1, e_2, \dots, e_{n-1},$$

with

$$e_1 = T = c'.$$

Choose e_n so that

$$(e_1, \dots, e_n)$$

is positively oriented. This is the Frenet frame of c .

For $n = 3$, it is precisely (T, N, B) .

Definition. The Frenet curvatures $\kappa_1, \dots, \kappa_{n-1}$ are defined by

$$\kappa_i = \langle e'_i, e_{i+1} \rangle, \quad 1 \leq i \leq n-1.$$

The Gram–Schmidt construction gives

$$\kappa_1, \dots, \kappa_{n-2} > 0.$$

The last curvature κ_{n-1} may have either sign.

In $\mathbb{R}^3$,

$$\kappa_1 = \kappa, \quad \kappa_2 = \tau.$$

The functions $\kappa_1, \dots, \kappa_{n-1}$ measure the successive ways in which the curve leaves the spaces spanned by its earlier derivatives.

The Frenet equations in $\mathbb{R}^n$ are

$$\begin{aligned} e'_1 &= \kappa_1 e_2, \\ e'_i &= -\kappa_{i-1} e_{i-1} + \kappa_i e_{i+1}, \quad 2 \leq i \leq n-1, \\ e'_n &= -\kappa_{n-1} e_{n-1}. \end{aligned}$$

If $\mathbf{e} = (e_1, \dots, e_n)^\top$, then

$$\mathbf{e}' = K(s)\mathbf{e},$$

where

$$K(s) = \begin{pmatrix} 0 & \kappa_1 & & & 0 \\ -\kappa_1 & 0 & \kappa_2 & & \\ & -\kappa_2 & 0 & \ddots & \\ & & \ddots & \ddots & \kappa_{n-1} \\ 0 & & & -\kappa_{n-1} & 0 \end{pmatrix}$$

The matrix $K(s)$ is skew-symmetric, reflecting the fact that the frame remains orthonormal.

Fundamental theorem of the local theory of curves.

Let I be an interval and let

$$\kappa_1, \dots, \kappa_{n-1} : I \longrightarrow \mathbb{R}$$

be smooth functions satisfying

$$\kappa_1, \dots, \kappa_{n-2} > 0.$$

Fix $s_0 \in I$, a point $p \in \mathbb{R}^n$, and a positively oriented orthonormal basis $(b_1, \dots, b_n)$ of $\mathbb{R}^n$.

Then there is a unique unit-speed Frenet curve

$$c : I \longrightarrow \mathbb{R}^n$$

such that

$$c(s_0) = p, \quad e_i(s_0) = b_i, \quad \text{and} \quad c \text{ has Frenet curvatures } \kappa_i.$$

Equivalently, the Frenet curvatures determine the curve uniquely up to an orientation-preserving Euclidean motion.

Idea of the proof.

First solve the linear system

$$\mathbf{e}' = K(s)\mathbf{e}, \quad \mathbf{e}(s_0) = (b_1, \dots, b_n)^\top.$$

Existence and uniqueness for linear differential equations gives a unique frame. Since K is skew-symmetric, differentiation shows that

$$\langle e_i, e_j \rangle$$

is constant, so the solution remains an orthonormal frame.

Then recover the curve by integrating its tangent:

$$c(s) = p + \int_{s_0}^s e_1(u) du.$$

Thus $c' = e_1$, so c has unit speed. The Frenet equations ensure that its Frenet curvatures are exactly the prescribed functions.

The geometry of the curve has been reduced to an initial-value problem for a system of differential equations.